



REPUBLIC OF TURKEY

GEBZE TECHNICAL UNIVERSITY

FACULTY OF ENGINEERING

PROJECT PROPOSAL

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Integrated ML and DL Architectures for Mobile Traffic Anomaly Detection

Project Objective

The objective of this project is to develop an end-to-end anomaly and intrusion detection system. The system will leverage **Machine Learning (ML) and Deep Learning (DL)** models to identify anomalous network traffic and potential cyber threats. To demonstrate operational feasibility, the project features a streaming dashboard that simulates network traffic analysis and triggers automated alerts.

Dataset and Methodology

The system will be trained and evaluated using a recognized network intrusion dataset, specifically either KDD or UNSW-NB15, depending on the final feature requirements.

Model Strategy and Benchmarking

The project will implement a multi-model approach to ensure high accuracy and reliability. We will train custom ML and DL models tailored to the dataset's specific feature set. These custom architectures will be rigorously evaluated against established, pre-trained benchmark models. All models will be housed in a centralized Model Repository, allowing the system to easily swap models for active inference and continuous benchmarking.

System Architecture Components

- **Simulation Engine:** A module that simulates a continuous data stream using a separate portion of the dataset not utilized during the training or testing phases. It outputs this data sequentially to mimic live packet flow in a mobile network environment.
- **Wrapper Engine:** A Python-based inference server that contains both the custom-trained ML/DL models and the pre-trained models used for benchmarking. It receives the streamed data, scales the features appropriately, and passes them through the active models. It strictly enforces safe software design principles; notably, the Model Repository is fully encapsulated, ensuring that any internal ID pointer is never shared with or exposed to other classes.
- **Streamlit Dashboard:** The frontend presentation layer connected to the backend, providing an updating UI divided into two primary functional areas:
 - **Monitoring and Alerting:** A centralized view displaying the continuous incoming network traffic stream alongside specific, automated visual alerts generated whenever an anomaly or intrusion is detected by the Wrapper Engine.
 - **Benchmarking Interface:** A dedicated page to present and compare the real-time and static evaluation metrics of the custom models against the pre-trained benchmarks.

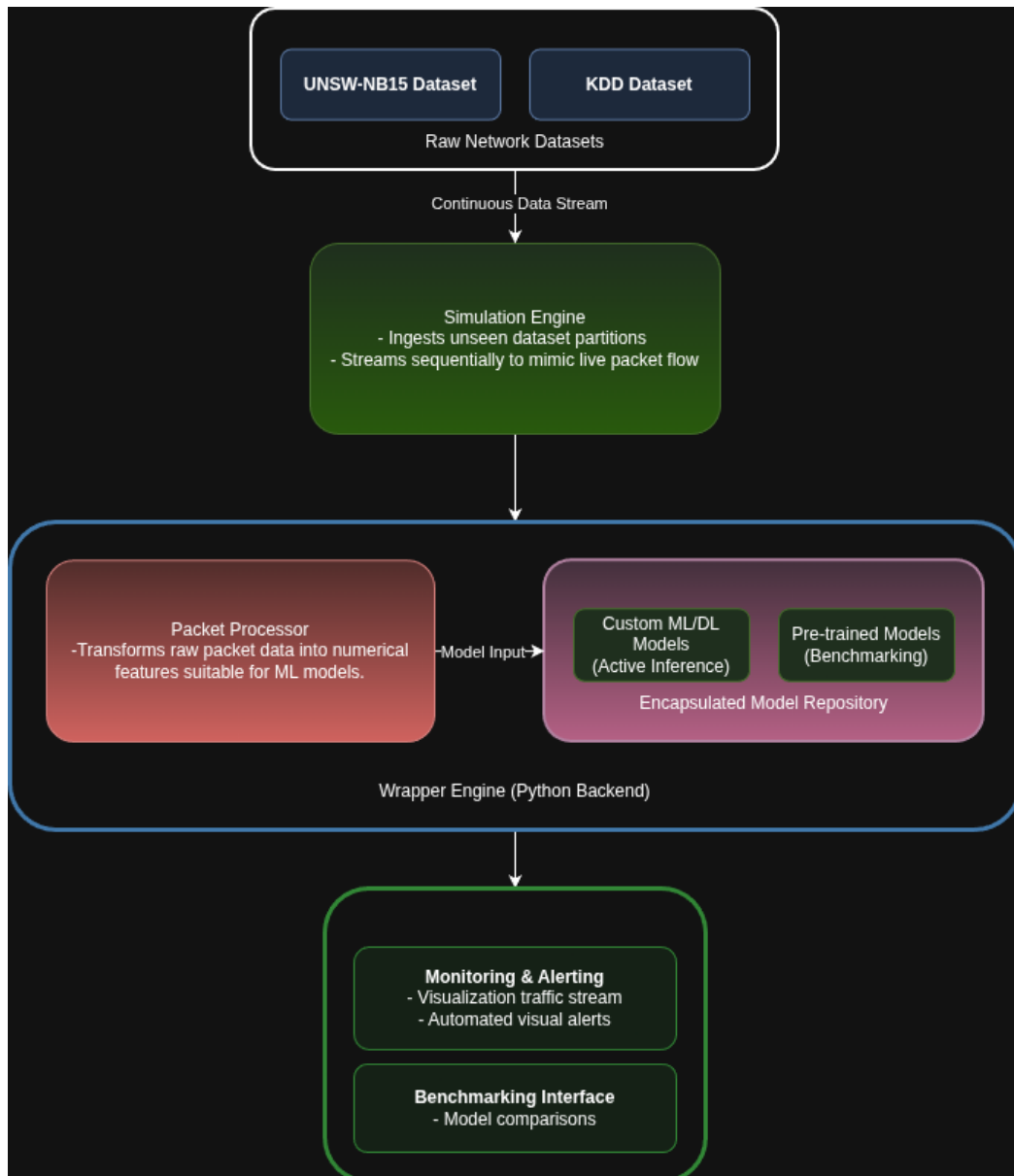


Figure 1. System architecture of the proposed anomaly and intrusion detection pipeline, illustrating the end-to-end data flow from raw dataset simulation through the encapsulated wrapper engine to the frontend visualization dashboard.

References

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